



A Curated Modlist & Guide for Assetto Corsa

Version 1.0 — July 2026

Compatible with Assetto Corsa v1.16.4 (August 2020)

*Prepared for the driving enthusiast who wants a curated, stable,
and progressively-enhancing modding experience — from stock to spectacular,
one wave at a time.*

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About This Guide

Assetto Maximus is a curated modlist and comprehensive game guide for **Assetto Corsa** (2014), version 1.16.4 (August 2020).

It is designed to take you from a completely vanilla installation to a fully-modded sim racing experience through progressive, carefully-tested waves.

Credits

- **Author:** The Assetto Maximus team
- **Mod Credits:** Each mod's respective author(s) — see individual mod pages for full credits
- **Special Thanks:** The Assetto Corsa modding community

Assetto Corsa is a product of Kunos Simulazioni. This guide is a community project and is not affiliated with or endorsed by Kunos Simulazioni.

Version

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Introduction

What is Assetto Maximus?

Assetto Maximus is a curated modlist and comprehensive game guide for **Assetto Corsa** (2014), version 1.16.4 (August 2020). It is designed to take you from a completely vanilla installation to a fully-modded sim racing experience through progressive, carefully-tested waves.

The project is built around three principles:

- **Stability first** — Every mod is verified for compatibility with AC v1.16.4. No conflicts, no broken builds.
- **Progressive complexity** — Each wave builds naturally on the previous one, introducing new mechanics and content only when you are ready for them.
- **No cheats** — No omniscient overlays, no physics exploits, no unfair advantages. Skill should be the limiting factor.

Wave Structure

Each wave pairs a curated modlist with a roleplaying narrative to give context to your progression:

Wave	Title	Theme
0	The Novice's Tale	Vanilla+ — learn the fundamentals
1	The Apprentice's Rise	Enhanced — deeper engagement

Future waves will be added as the guide evolves.

How to Use This Guide

1. **Start fresh** — Install Assetto Corsa from Steam (v1.16.4)
2. **Follow Wave 0** — Set up Content Manager, install UI/QoL mods, learn to drive
3. **Progress to later waves** — Add more mods as you gain experience
4. **Refer to appendices** — For troubleshooting, reference, and configuration details

Typographic Conventions

`Code / Command text` — Terminal commands, file paths, or keyboard shortcuts

Italic text — Emphasis or first mention of a term

[Blue underlined text](#) — Clickable links

Installation Guide

Assetto Corsa — Base Installation

1. Purchase and download **Assetto Corsa** on Steam
2. Right-click the title in your library → **Properties** → **Betas** → select 1.16.4 from the dropdown
3. Wait for the download to complete
4. Launch the game once to generate configuration files, then exit

Verifying Your Version

In the main menu, check the bottom-right corner for version text. It should read v1.16.4 (build date August 2020).

Content Manager

[Content Manager](#) (CM) is the essential launcher and mod manager for Assetto Corsa. It replaces the default launcher with a significantly more capable interface.

Download & Install

1. Visit [the official CM site](#)
2. Download the latest full release (not a patch)
3. Run the installer — it will detect your Assetto Corsa installation automatically
4. Launch Content Manager after installation

Initial Setup

1. Go to **Settings** → **Content Manager** → **Appearance**
 - Set **Theme** to your preference (Dark is recommended for readability)
 - Set **Language** if needed
2. Go to **Settings** → **Assetto Corsa**
 - Verify that the **Game path** points to your AC installation (e.g., C:\Program Files (x86)\Steam\steamapps\common\assetto corsa)
3. Go to **Settings** → **Customs**
 - **Cars search path**: Leave default
 - **Tracks search path**: Leave default

Installing Mods with CM

CM supports drag-and-drop installation:

- `.zip` / `.rar` / `.7z` archives containing car or track folders
- `.kn5` / `.kn` files (compiled Assetto Corsa data)
- Folders placed directly into `assetto corsa/content/cars` or `assetto corsa/content/tracks`

To install, simply drag the archive onto the CM window, or use **Install Mod** button.

Custom Shaders Patch (CSP)

Custom Shaders Patch is a major visual and functional enhancement for Assetto Corsa. It will be added in Wave 1.

Note for later: CSP versions must match AC v1.16.4. The last compatible pre-WeatherFX build is `0.1.79`. For later CSP versions with WeatherFX, use v1.16.4-post-launch compatible builds (`0.2.0` +).

Recommended Folder Structure

```
Steam/steamapps/common/assetto corsa/
├── content/
│   ├── cars/           - Car mods
│   ├── tracks/        - Track mods
│   └── fonts/         - Custom fonts (rare)
├── extensions/        - CSP configs
├── system/cfg/        - Game configuration files
└── assetto corsa.exe  - Game executable
```

Backup Your Installation

Before adding any mods, create a backup of your clean AC installation:

```
Copy "assetto corsa" folder to "assetto corsa-stock"
```

This will let you revert if anything goes wrong.

Wave 0: The Novice's Tale

"The checkered flag means nothing when you cannot find the racing line."

The Narrative

You have just passed your driving test. Your budget is modest — a compact hatchback, a second-hand set of tires, and a fire in your chest that matches the red of the setting sun. You sign up for your first track day with nothing but a helmet, a full tank, and the will to learn.

You do not need a faster car. You need to **become** faster.

Wave 0 Philosophy

Wave 0 is a **Vanilla+** experience. No game-changing mods. No physics overhauls. No content packs. The goal is to learn Assetto Corsa at its purest while improving the UI/UX to modern standards.

- **What changes:** Interface, HUD, quality of life
- **What stays stock:** Physics, content, graphics (as shipped)

Understanding the HUD

The default Assetto Corsa HUD displays:

- **Speedometer & Tachometer** — Upper corners
- **Gear indicator** — Center bottom
- **Lap timer** — Top center
- **Relative display** — Left side (shows nearby cars)
- **Rearview mirror** — Toggle with F2

Controls

Keyboard & Mouse (Basic)

Action	Default Key
Steer	A / D
Accelerate	W

Action	Default Key
Brake	S
Gear up / down	Shift / Ctrl
Look left / right	Left Arrow / Right Arrow
Look back	Down Arrow
Rearview mirror	F2
Toggle HUD	Ctrl + H
Change camera	F1 (cycle)
Pause	Esc

Gamepad

Action	Xbox	PlayStation
Steer	Left stick	Left stick
Accelerate	RT	R2
Brake	LT	L2
Gear up / down	A / X	× / □
Clutch	LB	L1
Look left / right	Right stick	Right stick
Look back	Right stick click	R3
Rearview mirror	Back	Share
Toggle HUD	Guide + H	PS + H
Change camera	Y	△
Pause	Menu	Options

Tips for gamepad:

- Set **Steering Speed** to 50—70% in Content Manager controls for linear response
- Enable **Steering Filter** at low values (0—5) to smooth input
- Increase **Deadzone** to 5—10% if the stick drifts
- Use **Vibration** (enabled in Settings → Controls) for tyre slip feedback
- For throttle control: squeeze RT / R2 gradually — abrupt application causes spin

Wheel Users

Configure your wheel in **Controls** → **Axis**. Key settings:

- **Steering lock**: Match your wheel's physical rotation (e.g., 900°)
- **Gamma**: 1.00 (linear response)
- **Filter**: 0 (no smoothing)
- **Force Feedback**: 100% gain (adjust per car)

See Configuration chapter for detailed FFB tuning.

Driving Curriculum: Phase 1 — Foundations

Goal: Stop crashing and build clean, repeatable laps.

1.1 Seating Position & Ergonomics

Before you turn a wheel, set up your physical environment. A correct seating position is the single cheapest performance gain available.

Wheel Position

- Sit so your shoulders remain relaxed against the seat back when your hands are at quarter-to-three
- Your elbows should be bent at 90—120°
- You should be able to turn the wheel full lock without leaning forward or stretching

Pedal Placement

- Your heel should rest on the floor, allowing your foot to pivot from the ankle
- The brake pedal should be reachable without lifting your heel
- For heel-toe downshifting: the throttle and brake should be close enough in height to blip with your right foot's edge

Field of View (FOV)

FOV is the single most important visual setting for accuracy. Too wide and distances compress (you brake too early). Too narrow and you lose peripheral awareness.

In ideal FOV: the dashboard width matches what you see in real life when sitting in the car.

To calculate your ideal FOV:

1. Measure your monitor width (in cm) and distance from your eyes to the screen
2. Use an online FOV calculator (set result as "Vertical FOV" in AC)
3. Or start with 35—45° (single monitor) and adjust until braking points feel natural

1.2 Vision Fundamentals

Your eyes lead the car. A common beginner mistake is staring at the front bumper or the track directly ahead.

Where to Look

- **Entry:** Look at the braking point, then the turn-in point
- **Mid-corner:** Look at the apex
- **Exit:** Look at the exit curb or track-out point
- **Between corners:** Look as far ahead as possible — ideally 2—3 turns ahead

Reference Points

Choose fixed objects (braking marker boards, curb colors, marshall posts) as reference points. Without reference points every lap is a guess; with them, you can measure and improve.

- **Brake marker:** "I brake when my front bumper passes the 100m board"
- **Turn-in:** "I turn when the curb pattern changes from red to white"

- **Apex:** “I clip the apex where the outer barrier has the yellow stripe”

1.3 Smooth Inputs

Jerky inputs upset the car’s balance. Every control movement should be gradual and deliberate.

Throttle

- Squeeze, do not stomp. Apply throttle progressively over about 0.3—0.5 seconds
- On exit, increase pressure as the steering straightens
- Abrupt lift-off mid-corner causes snap oversteer

Brake

- Apply pressure smoothly (not a stomp), building to maximum in about 0.1—0.2 seconds
- Release pressure gradually as you approach turn-in
- A sudden brake release unweights the front and reduces turn-in grip

Steering

- Avoid sawing at the wheel. Plan your steering input before you execute it
- Smooth, continuous rotations — not a series of small corrections
- If you need multiple corrections mid-corner, you turned in too early or too late

1.4 Understanding Grip Limits

Grip is finite. Every tyre has a maximum force it can deliver. Your job as a driver is to stay under that limit — and sense when you are near it.

The grip circle (or friction circle) concept:

- A tyre can deliver 100% grip total
- If you use 80% for braking and 80% for turning, you exceed 100% — the tyre slides
- **Traction pyramid:** Brake in a straight line, then turn, then accelerate. Never do two things at full effort simultaneously

Learn to feel the limit in practice: drive a circle in low gear and slowly increase speed until the car slides. This is the limit. Back off slightly — that is where speed lives.

1.5 The Racing Line

The fastest path through any corner follows the **outside-inside-outside** principle:

1. **Brake zone:** Stay wide on the outside — sets up a straighter braking zone
2. **Turn-in:** Aim for the apex (inside curb)
3. **Exit:** Let the car drift wide to the outside curb

Corner Types

- **90° turn:** Standard outside-inside-outside
- **Hairpin:** Very late apex to open up the exit
- **Sweeper:** Early apex, carry as much speed as possible
- **Chicane:** Connected left-right — sacrifice entry of the first to set up the second

1.6 Threshold Braking

Braking at the absolute limit of grip — the point just before the wheels lock — is called threshold braking.

Technique

1. Apply brake firmly but progressively
2. Listen to tyre squeal and feel pedal vibration (ABS off) or slight pedal pulsation (ABS on)
3. Reduce pressure slightly if you feel a wheel lock
4. The goal: maximum deceleration without locking

Without ABS

Most road cars in AC do not have ABS. Practice in the Mazda MX-5 Cup (ABS off):

- Apply brakes to about 80% initially
- Increase to near-lock as you feel the grip available
- If the front wheels lock, you cannot steer — release slightly to regain steering authority

1.7 Corner Phases

Every corner has three phases. Master each separately, then join them.

Entry (Brake Zone → Turn-in)

- Brake in a straight line
- Release brakes progressively
- Turn the wheel smoothly toward the apex

Mid-Corner (Apex)

- Steady throttle or coasting
- Minimum speed occurs here — feel for it
- Car should be settled, not sliding

Exit (Apex → Track-Out)

- Progressive throttle application
- Let the steering straighten naturally
- Track out to the outside curb

1.8 Recognizing Understeer vs Oversteer

Understeer (Push)

- **Feeling:** The front washes wide; you turn more but the car does not respond
- **Cause:** Too much speed at entry, or too much steering angle for the grip available
- **Fix:** Reduce entry speed, trail brake slightly, or add more rear brake bias

Oversteer (Loose)

- **Feeling:** The rear steps out; the car tries to spin

- **Cause:** Too much throttle mid-corner, abrupt lift-off, or too much entry trail braking
- **Fix:** Smooth throttle application, reduce trail braking depth, adjust rear wing or ARB

Learn to catch oversteer with steering correction (countersteer) — this will save you hundreds of times.

1.9 Building Consistent Lap Times

Clean laps are faster than spectacular laps. A spin costs 3—5 seconds. A four-wheel-off costs 1—2 seconds. A perfect lap with one mistake loses to a clean lap every time.

The Consistency Method

1. Run 10 laps without counting your time
2. Focus on hitting your reference points every lap
3. Once you can hit them 8/10 times, start timing
4. Your goal: 5 consecutive laps within 0.5 seconds of each other
5. **Then** push for outright speed

Practice Focus

- **Car:** Mazda MX-5 Cup or Abarth 500 (simple, forgiving)
- **Tracks:** Magione, Brands Hatch Indy, Monza Junior, Imola short layouts
- **Session:** 20—30 minute practice sessions — short enough to stay focused, long enough to build patterns

Wave 0 Modlist

Wave 0 includes only UI, HUD, and quality-of-life mods. No mechanics, graphics, or content changes.

UI & Quality of Life

- [Sidekick](#) — Audible spotter for proximity, gaps, and flags. **Dependencies:** Content Manager. **Impact:** Audio notification system, no gameplay change.
- [Car Radar](#) — Visual radar display showing nearby cars. **Dependencies:** Content Manager. **Impact:** Situational awareness tool, no physics change.
- [Helicorsa](#) — 3D arrow indicator showing car proximity and relative position. **Dependencies:** Content Manager. **Impact:** Visual overlay, no gameplay change.
- [Crew Chief](#) — External spotter and race engineer application. **Dependencies:** Standalone app (runs alongside AC). **Impact:** Voice-based spotter, pit strategy assistance, no in-game modification.

Installation Instructions

1. Download each mod from the link provided
2. Drag the `.zip` / `.rar` onto Content Manager
3. CM handles extraction and installation automatically
4. Enable each mod in CM's **Mods** tab
5. Configure in-game overlays via CM's **Settings** → **Assetto Corsa** → **Apps**

Verification

After installing Wave 0, launch AC through Content Manager. You should see:

- The Sidekick spotter reads out gaps and flags
- Car Radar shows a 360° proximity display
- Helicorsa highlights nearby cars with arrows
- Everything else is identical to stock Assetto Corsa

Wave 1: The Apprentice's Rise

"Speed is a byproduct of control. Learn control first."

The Narrative

Months of track days have paid off. Your lap times are consistent, your car control is instinctive, and you have caught the attention of a local club racing organizer. They offer you a seat in their amateur series.

Your hatchback is retired. A purpose-built track car sits in the garage. You have graduated from **driving** — now you learn to drive **fast**.

Wave 1 Philosophy

Wave 1 builds on the solid foundation of Wave 0. UI/QoL mods remain. Now we add:

- **Graphics enhancements** — CSP, weather, lighting
- **Content** — New cars and tracks
- **Practice infrastructure** — Telemetry tools, FFB optimization, data logging

Driving Curriculum: Phase 2 — Balance & Speed

Goal: Carry more speed by understanding weight transfer and car balance.

2.1 Weight Transfer Fundamentals

Everything in high-performance driving flows from weight transfer. Understanding where the weight is tells you where the grip is.

Under Braking

- Weight shifts **forward** — front tyres gain grip, rear tyres lose grip
- The car becomes more willing to turn (more front grip)
- The rear becomes less stable — lift-off oversteer risk is highest here

Under Acceleration

- Weight shifts **backward** — rear tyres gain grip, front tyres lose grip
- The car becomes less willing to turn (understeer on power)
- Traction control helps, but smooth throttle is better

In Transition

- Left-to-right weight transfer happens at turn-in
- The rate of transfer depends on how quickly you turn the wheel
- Aggressive steering = rapid transfer = potential spin

The **weight transfer pendulum**: Brake to load the front → turn in while releasing brake → the front has maximum grip exactly at turn-in. This is why releasing the brake and turning must overlap.

2.2 Trail Braking

Trail braking means carrying brake pressure **into** the corner, not releasing it all before turn-in. This keeps weight on the front tyres to improve rotation.

How to Practice Trail Braking

1. Brake in a straight line as normal
2. At the turn-in point, instead of fully releasing the brake, keep 10—20% pressure
3. As you turn the wheel, **gradually** release the remaining brake
4. The brake should reach zero at or just before the apex

Common Mistakes

- **Too much brake at turn-in** — This causes the rear to step out (spin)
- **Releasing too fast** — The car understeers because the front unweights suddenly
- **Trailing too deep** — Braking past the apex costs exit speed

2.3 Brake Release Timing & Modulation

How you release the brake is as important as when you apply it.

- **Linear release**: Steady, even pedal release — good for high-speed corners
- **Quick initial release, then slow**: Use when you need to stabilize the car before turning
- **Slow initial release, then quick**: Use when you want maximum front grip for a late turn-in

The brake release profile should match the corner. Experiment with different release shapes on the same corner and watch the lap time.

2.4 Throttle Application for Rotation & Traction

The throttle is not just for going faster — it is a car control tool.

Power-On Rotation

Applying throttle mid-corner shifts weight to the rear. In a rear-wheel-drive car, this can **tighten** the line (reduce understeer) by inducing slight oversteer at the rear. This is called **power oversteer** or **throttle steering**.

In a front-wheel-drive car, power-on causes **understeer** (the front washes wide). In this case, **lifting** the throttle shifts weight forward and tightens the line.

The Throttle Shapes

- **Squeeze**: Gradual increase from apex to exit — safe, predictable

- **Punch:** Early sharp application to rotate the car, then back off — advanced technique
- **Blip:** Quick throttle for downshift rev-matching (mandatory for non-sequential cars)

2.5 Slip Angle Awareness

The fastest way through a corner is not at zero slip angle (perfect grip) — it is at a small, controlled slip angle. The rear of the car should be working slightly to rotate.

The ideal slip angle is 2—6° of rear yaw. You can feel it as a gentle looseness — the car is “working” but not sliding.

To find it: drive a constant-radius corner in second gear. Add throttle until you feel the rear start to move. Reduce throttle slightly. That position — right at the edge of the slide — is where the car rotates fastest.

2.6 Managing Understeer & Oversteer Actively

In Phase 1 you learned to **recognize** understeer and oversteer. Now learn to **manage** them.

Correcting Understeer Mid-Corner

1. **Reduce steering angle** — Turning more makes it worse (front tyres are already at the limit)
2. **Slightly lift or hold throttle** — This transfers weight forward, loading the front tyres
3. **Or brake gently** — Trail-brake to force rotation

Correcting Oversteer Mid-Corner

1. **Countersteer immediately** — Point the front wheels where you want to go
2. **Lift or reduce throttle** — Treading carefully: complete lift can snap the car back the other way
3. **Catch and hold** — Maintain countersteer until the car stabilizes, then smoothly straighten

2.7 Minimum Corner Speed

The slowest point in any corner occurs at or just after the apex. Knowing where minimum speed occurs and what affects it is central to fast driving.

- **If the minimum speed is too low:** You are braking too deep, not trail-braking enough, or over-slowng
- **If the minimum speed is too high:** You will run wide on exit or lose traction on power
- **Target:** Minimum speed should occur at the apex, not before it

To check: use a telemetry trace and find the speed graph. The valley should align with the apex marker.

2.8 Using Kerbs Correctly

Kerbs are not obstacles — they are tools. But each type of kerb behaves differently.

- **Flat red/white kerbs:** Safe to drive on. Use them to extend the track width
- **Tall “sausage” kerbs:** Will upset the car. Avoid them except at very low speed
- **Painted kerbs (raised bumps):** Can unsettle the car. Two wheels on is fine; four wheels is risky

Kerb Technique

- **Entry kerb:** Two wheels on kerb to open up the corner
- **Apex kerb:** Clip the inside edge — do not let the car jump
- **Exit kerb:** Two wheels on to maximize track width on exit

2.9 Basic Telemetry Reading

Now that you have data logging tools (see Wave 1 modlist), you can compare laps.

The Three Traces

1. **Brake trace:** Should show a smooth, quick application followed by a controlled release. Any stuttering means inconsistent pedal work
2. **Throttle trace:** Should be smooth on application. An “on-off-on-off” pattern reveals nervous driving
3. **Steering trace:** Should show one smooth arc into the corner and one smooth arc out. Multiple corrections indicate poor entry planning

Comparing Laps

Overlay a reference lap (your best or a faster driver’s) against a normal lap:

- Where is the brake trace different? (Braking earlier or later?)
- Where is the minimum speed different? (Carrying more or less speed?)
- Where is the throttle-on point different? (Getting on power sooner or later?)

Practice Focus: Phase 2

- **Cars:** Mazda MX-5 Cup, Toyota Supra MKIV, Porsche 911 GT3 RS
- **Tracks:** Imola, Brands Hatch (full), Suzuka — mid-tier technical tracks
- **Goal:** Trail-brake into every corner for a full session. You will spin — that is learning.

Driving Curriculum: Phase 3 — Cornering Precision

Goal: Extract maximum from every corner type.

3.1 Advanced Trail Braking & Rotation

Phase 2 introduced trail braking. Phase 3 refines it.

Late Trail Braking

On high-speed corners where you need maximum rotation, carry brake pressure deeper into the corner — past turn-in, toward the apex. This **forces** the car to rotate.

This is high-risk, high-reward. Too much and you spin. Too little and you understeer. The margin is narrow.

Rotation Points

Every car has a characteristic rotation point — the moment in the corner where the rear begins to rotate toward the apex. Identifying this point consciously lets you manipulate it:

- **Earlier rotation:** Trail brake deeper, turn in more sharply
- **Later rotation:** Release brake earlier, turn in more gradually

3.2 Spiral / Progressive Steering

Instead of turning the wheel to a fixed angle, use a **progressive** steering technique: increasing steering angle as you approach the apex, then decreasing on exit.

This creates a spiral path — the tightest radius at the apex, wider before and after.

How to Practice

1. On a long sweeper (e.g., Parabolica at Monza), enter with very little steering angle
2. **Gradually** add steering angle as the corner progresses
3. The maximum steering angle should coincide with the apex
4. On exit, gradually reduce steering

A constant steering angle through a corner means you are leaving time on the table — the car can turn tighter at the apex than at entry or exit.

3.3 Identifying Rotation Points

Your rotation point is where the car's yaw (rotation) matches the corner radius. Before this point, the car is still "straight." After this point, the car rotates toward the apex.

To find it: on corner entry, pay attention to when the car starts to face the apex. **That feeling** is your rotation point. Once you can feel it, you can control it with brake and throttle.

3.4 Double Apexes & Complex Corners

Some corners require two apexes because the road is not a simple arc.

Double Apex Technique

1. Clip the first apex on entry
2. Let the car drift slightly wide between apexes
3. Clip the second apex at mid-corner
4. Exit normally

Key insight: Do not try to carry a single line through a double-apex corner. Treat it as two connected corners — sacrifice a little between them to nail both apexes.

Examples: Suzuka's Spoon Curve, Nürburgring's Hatzenbach

3.5 Blind Corners & Commitment

Blind corners — where you cannot see the exit — require faith in your reference points. If you set your braking point and turn-in correctly, the car will be on the correct line even if you cannot see where it is going.

Rule: If you are not a little scared in a blind corner, you are not at the limit. Trust your marks, not your eyes.

Practice

Choose a blind corner (e.g., Eau Rouge at Spa, or the dip before the final corner at Bathurst). Cover the top half of your screen with tape. Drive the corner using only your reference points.

3.6 Elevation Changes & Camber Effects

Elevation

- **Uphill braking:** Less grip uphill because weight transfers rearward. Brake earlier
- **Downhill braking:** More grip downhill because weight loads the front. Brake later but more carefully — the car is unstable
- **Uphill corners:** Natural slowing effect — you can carry more speed
- **Downhill corners:** Natural speeding effect — you need to brake earlier

Camber

- **Negative camber (road tilts into corner):** More available grip. Carry more speed
- **Positive camber (road tilts away from corner):** Less grip. The car will want to run wide

Read the road: look at the surface angle on approach. If the road falls away, reduce speed. If it banks, trust it.

3.7 Compound Corners (Linking Multiple Turns)

Compound corners are sequences where the exit of one turn feeds directly into the entry of the next (e.g., the S's at Suzuka or Maggotts-Becketts-Chapel at Silverstone).

The Secret

You cannot take the optimal line through both corners. You must **compromise**:

- **First corner:** Take a line that sets up a good entry for the second
- **Second corner (or last in sequence):** Take the optimal line — this is where you get the time back

General rule: sacrifice entry of the first to maximize exit of the last.

3.8 Line Variation

In Phase 1 you learned the single racing line. Now learn to vary it.

Defensive Line

- Take the inside of the corner
- This sacrifices exit speed but protects the position
- Use when a car is alongside or behind

Attack Line

- Take the standard racing line (outside → apex → track-out)
- Maximizes exit speed
- Use when clear ahead or trying to pass on the next straight

Compromise Line

- A middle-ground line that preserves some exit speed while covering the inside
- Use in close racing where you do not know which side the attacker will take

3.9 Tyre Temperature & Pressure Management

At this level, you need to read the tyres — not just react to them.

Temperature Profile

- **Inner edge hotter than outer:** Too much negative camber — the car lacks straight-line braking grip
- **Outer edge hotter than inner:** Too little negative camber — the car lacks cornering grip
- **Center hotter than edges:** Over-inflated — reduced contact patch
- **Edges hotter than center:** Under-inflated — wallowing, poor response

Managing Wear

- Hot tyres lose grip. After 2—3 fast laps, the peak grip window closes
- In a hot lap: push on the first flying lap (peak grip)
- In a long race: manage pace to save tyres for the final stint

Practice Focus: Phase 3

- **Cars:** Porsche 911 GT3 RS, Ferrari 488 GTE, progressively faster machinery
- **Tracks:** Spa, Suzuka, Nürburgring GP, Laguna Seca — long circuits with diverse corner types
- **Goal:** Practice double-apex and compound corners specifically. Pick one complex sequence and run it 50 times

Tuning Reference

Tyre Pressures

- **Target range:** 26—28 PSI when hot
- **Too low:** Car feels sluggish, tyres overheat quickly
- **Too high:** Car feels twitchy, reduced contact patch
- Check pressures after 3—4 laps of consistent driving

Camber

- Negative camber improves cornering grip at the expense of straight-line braking
- **Start with:** -2.5° front, -1.5° rear
- **Adjust:** If the car understeers at corner entry, add more front negative camber

Anti-Roll Bars (ARB)

- Softer bar = more mechanical grip, more body roll

- Stiffer bar = less grip transfer, quicker response
- **Understeer fix:** Soften front ARB or stiffen rear ARB
- **Oversteer fix:** Stiffen front ARB or soften rear ARB

Wave 1 Modlist

Essential Tools

- [Custom Shaders Patch \(CSP\)](#) — Advanced graphics, weather, and post-processing. **Dependencies:** Content Manager. **Impact:** Major visual overhaul, adds weather/lighting system. Requires compatible version for AC v1.16.4.
- [Sol](#) — Weather simulation and lighting system (requires CSP). **Dependencies:** Custom Shaders Patch. **Impact:** Dynamic weather, time-of-day cycle, realistic lighting.

Graphics Mods

Content Mods (Cars)

- [Ferrari 488 GTE](#) — GTE/GTLM-class endurance racer. **Dependencies:** None.

Content Mods (Tracks)

- [Mount Panorama Bathurst](#) — Legendary Australian circuit. **Dependencies:** None.
- [Nürburgring Nordschleife](#) — The Green Hell. **Dependencies:** None.
- [Spa-Francorchamps](#) — 2022 layout. **Dependencies:** None.
- [Brands Hatch](#) — Classic British circuit. **Dependencies:** None.

Mechanics Mods

- [FFB Clip](#) — Force feedback optimization tool. **Dependencies:** Content Manager. **Impact:** Prevents FFB clipping, improves wheel feel. **System/mechanic:** Force feedback system enhancement.
- [Minimalista HUD](#) — Clean data HUD replacement. **Dependencies:** Content Manager. **Impact:** Dashboard overlay with telemetry data. Replaces old RST Minimal UI.

Optional

- [3D Shift Light](#) — Visual gear shift indicator. **Dependencies:** Content Manager.

Installation Sequence

1. Ensure Wave 0 is fully installed and verified
2. Install **Custom Shaders Patch** via Content Manager
3. Install **Sol** (requires CSP active)
4. Install all other Wave 1 mods in any order

5. Verify CSP is active

Wave 2: The Master's Ascent

"When you are no longer fighting the car, you can fight the race."

The Narrative

You have the speed. You have the consistency. Now you need to **win**.

A professional coaching program has taken notice of your results. You are invited to an elite driver development camp — a week of advanced techniques, race simulations, and mental conditioning. The other drivers are just as fast as you. The margin between victory and defeat will not come from raw pace.

It will come from craft.

Wave 2 Philosophy

Wave 2 completes the Assetto Maximus Driving Curriculum. All previous mods remain. New additions focus on **racing infrastructure**:

- **Advanced telemetry & data analysis** — Full MoTeC-style lap comparison
- **Weather & dynamic conditions** — Sol/Pure weather scripts
- **High-fidelity content** — Premium car and track mods (if desired)
- **AI & race practice tools** — AI enhancement, grid starts

Driving Curriculum: Phase 4 — Race Craft & Mastery

Goal: Race intelligently and drive at the limit consistently.

4.1 Overtaking Techniques

The Setup

Overtakes are won before the braking zone. The actual pass is just the execution.

1. **Close the gap** — Use slipstream on the preceding straight to pull alongside
2. **Choose your side** — Commit to inside or outside before the braking zone
3. **Claim the corner** — You must be at least door-to-door before the apex

The Slipstream Pass

Follow within 0.3—0.5 seconds on a straight. The drag reduction gives you 5—10 km/h extra at corner entry. Brake slightly later and roll the inside line.

The Switchback

Fake an inside move to force the defender to cover the inside. At the last moment, change to the outside line. The defender, having committed to the inside, will have a slower exit — you carry more speed on the outside.

The Late Brake

Brake later than the defender, but **control** that brake — a dive-bomb that misses the apex and bumps the other car is not a pass; it is a crash. You must make the corner on the racing line, even if you are side-by-side.

4.2 Defensive Driving

The Single Move Rule

You may make one defensive move per braking zone. Once you choose your line, you must hold it. Weaving or moving twice is illegal and dangerous.

Inside Defense

Cover the inside line before the braking zone. This forces the attacker to go around the outside. You sacrifice exit speed (slower line through the corner) but protect the position.

Outside Defense

Stay on the racing line — do not cover the inside. This invites the attacker to try the inside, where they have to brake earlier. On exit, you have more speed and can pull away.

When to Yield

If an attacker has significant overlap before the apex, you must leave them space. Fighting a lost position often loses two positions — yield cleanly, fall in behind, and look for a counter-attack on the next straight.

4.3 Traffic Management & Awareness

Approaching Traffic

When catching slower cars (lapping):

1. Do not assume they see you — use your spotter (Sidekick / Crew Chief)
2. Choose a safe passing zone — ideally a straight with a long braking zone
3. **Point-by:** At the passing zone, point to the side you want them to let you through
4. Make the pass clean and immediate — do not race them

Being Lapped

1. Hold your line — do not weave or brake unpredictably
2. Lift slightly on the straight to let the faster car by

3. Stay on the racing line and let them choose the passing side
4. Once passed, fall in behind — do not try to re-pass

Multi-Class Awareness

In mixed-class racing (e.g., GT3 + GTE + slower classes):

- Faster classes are responsible for safe overtaking
- Slower classes should hold their line and be predictable
- Use your radar (Car Radar) and spotter to track closing speed differentials

4.4 Wet Weather Driving

The Wet Line

In the dry, you want the racing line (maximum rubber = maximum grip). In the wet, the racing line is **slippery** — rubber becomes grease when wet.

- **Avoid** painted lines, rubber buildup, and polished asphalt
- **Aim for** concrete patches, curb edges (where water drains), and less-traveled lines
- Braking zones shift: brake on areas with better drainage

Technique Adjustments

- **Brake earlier and smoother** — Threshold braking is dangerous in the wet
- **Feather the throttle** — Power oversteer happens at much lower speeds
- **Look further ahead** — Grip changes are easier to spot if you scan 5—10 seconds ahead
- **Increase following distance** — Spray reduces visibility to zero

Wet Line Variation

In a heavy rain race, the optimal line changes every lap as the track evolves. The driver who adapts fastest wins. Practice this:

1. Start a race on a dry track with Sol weather progression (time accelerated)
2. As rain begins, consciously search for grip on alternative lines
3. Note where standing water forms on the track — these become permanent danger zones

4.5 Drivetrain Characteristics

Each drivetrain behaves differently at the limit. Adapt your technique accordingly.

Front-Wheel Drive (FF)

- **Understeer is the default** — Throttle on corner exit pushes the nose wide
- **Lift to rotate** — Lifting the throttle shifts weight forward, helping turn-in
- **No power oversteer** — The front tyres do both the turning and the driving
- **Trail brake aggressively** — The rear is light; trail braking helps rotation

Rear-Wheel Drive (FR / MR)

- **Oversteer is available** — Power oversteer on exit, lift-off oversteer on entry
- **Throttle steering** — Use the throttle to tighten or widen your line mid-corner
- **Trail brake carefully** — Too much trail brake and the rear steps out

- **Smooth throttle essential** — Abrupt application = spin

All-Wheel Drive (AWD)

- **High entry grip** — AWD understeers less on entry than RWD
- **Power-on neutrality** — Add throttle mid-corner without spinning the rears
- **Understeer on exit** — The front tyres push when you apply power early
- **Brake later** — AWD allows later braking due to engine braking on all four wheels

Mid-Engine Rear-Wheel Drive (MR)

- **Extreme rotation** — The engine mass is behind the driver; turn-in is very sharp
- **Snap oversteer** — If the rear breaks loose, it happens fast — countersteer immediately
- **Throttle-sensitive** — Small changes in throttle position have large effects on balance
- **Smooth, smooth, smooth** — MR cars punish aggressive inputs

4.6 Class-Specific Techniques

GT3 / GTE

- ABS and traction control are available but should be set to minimum intervention
- Use trail braking heavily — these cars rotate well under brakes
- Kerb usage is aggressive — flat curbs are part of the racing line

Formula / Open-Wheel

- No ABS, no TC — pure driver input
- Braking is later and harder — high downforce means huge grip under brakes
- Steering input must be minimal — too much steering angle creates drag
- Downforce dependency: these cars have less grip at low speed, more at high speed

Road Cars / Touring Cars

- Minimal aero — mechanical grip is everything
- Slower corner entry, earlier power application
- Body roll is your feedback — feel it through the seat, not the wheel
- Mismatched tyre temps are a bigger issue than in downforce cars

4.7 Brake Bias Adjustment While Driving

Brake bias controls how much braking force goes to the front vs rear wheels. Adjusting it mid-race is an advanced skill that lets you adapt to changing conditions.

When to Change Bias

- **More rear bias (move bias backward):** When front tyres are overheating, or when the car understeers at corner entry
- **More front bias (move bias forward):** When the car is unstable under braking, or at high-speed tracks with long braking zones
- **Rain / low grip:** Move bias rearward to prevent front lock-ups
- **Low fuel (end of race):** Light car = less front grip. Move bias rearward

How Much

Adjust in 1—2% increments. A 5% change can transform the car's behavior.

4.8 Active vs Passive Driving Style

Passive Driving

Letting the car do the work. Minimal steering correction, predictable inputs, smooth transitions. This is fast when the car is well-balanced and the tyres are fresh.

Active Driving

Constantly managing the car's balance with brake, throttle, and steering corrections. The driver **creates** rotation, manages slip angle, and adjusts the line to track conditions. This is necessary when:

- The car is not perfectly balanced (always)
- Tyres are degrading
- Track conditions are changing (temperature, grip levels, rain line)
- Fighting for position

Great drivers are active, not passive. They are always working.

Mental State: Flow vs Overdrive

- **Flow:** The car and driver are in sync. Decisions happen without conscious thought. Laps feel effortless. This is the target state
- **Overdrive:** Trying too hard. Inputs become jerky. The driver is fighting the car. Lap times get worse, not better

If you find yourself overdriving: slow down. Do 3 laps at 80% effort. Reset your rhythm. Then build back up.

4.9 Mental Game

Focus Management

- A race is 30—60 minutes of sustained concentration. Your brain will fatigue.
- **Before the race:** Visualize the first 3 corners and the first lap
- **During the race:** Break the race into segments (quarters, or chunks of 5 laps)
- **Mistake recovery:** The moment after a mistake is the most dangerous — you are tempted to overcompensate. Reset: take the next corner at 90% to regain composure

Pressure Handling

- If a driver is pressuring you from behind: focus on the corner ahead, not the mirror
- If you are pressuring a driver ahead: be patient. The pass does not need to happen in the next corner. The race is long
- **Countdown method:** "I will attack for 5 corners. If I do not pass, I will drop back, cool the tyres, and try again in 2 laps"

Recovery from Mistakes

A spin, a lock-up, or an off-track moment does not end the race — unless you let it affect the next 10 corners.

1. Acknowledge the mistake
2. Tell yourself: "That lap is gone. The next one starts now."
3. Do one gentle corner to reset your rhythm
4. Build back to full attack over 2—3 corners

4.10 Full Telemetry Analysis & Self-Coaching

You have the tools (see Wave 2 modlist). Now use them to coach yourself.

Session Workflow

1. **Drive 5—10 clean laps** — Do not look at the timer
2. **Identify your best lap** — By feel, then confirm with lap time
3. **Overlay all laps** — Find the corners where your best lap gained time
4. **Compare to a reference** — Use a faster driver's lap (or your own target lap)
5. **Focus on one corner** — Do not try to improve every corner at once

What to Look For

- **Speed trace:** Where is my minimum speed lower than the reference?
- **Brake trace:** Am I braking earlier or later? Am I releasing smoothly?
- **Gear trace:** Am I in the right gear through the corners? Is my shift point optimal?
- **Steering trace:** Is there a "flat spot" where I hold constant steering (understeer)?

Self-Coaching Questions

- "Where am I losing the most time?"
- "Is this a confidence issue (I could carry more speed but I am afraid) or a skill issue (I do not know how to carry more speed here)?"
- "If I could change one thing about my driving today, what would it be?"

4.11 Setup Influence on Driving Technique

At this level, your driving technique should inform your setup, not the other way around.

- **If the car understeers:** Try adjusting trail braking depth before changing the front ARB
- **If the car is unstable on exit:** Change your throttle application technique before softening the rear ARB
- **If the car does not rotate on entry:** Experiment with brake bias (more rear) before changing the suspension

The rule: **Drive around the problem first. If you cannot fix it with technique in 5 laps, then change the setup.**

Practice Focus: Phase 4

- **Cars:** Ferrari 488 GTE (GT3/GTE), any Formula car (open-wheel), mix of drivetrains
- **Tracks:** Nürburgring Nordschleife, Spa, Bathurst — challenging circuits with varied corner types
- **Conditions:** AI races (higher aggression), online practice, mixed weather (Sol)
- **Goal:** Race the AI with 95% aggression for 15+ lap stints. Then review telemetry. Then race again

The Complete Curriculum Path

For reference, here is the full Assetto Maximus Driving Curriculum:

Phase	Title	Wave	Focus
1	Foundations	Wave 0	Consistency, confidence, basic car control
2	Balance & Speed	Wave 1	Weight transfer, trail braking, slip angle
3	Cornering Precision	Wave 1	Advanced techniques, complex corners
4	Race Craft & Mastery	Wave 2	Overtaking, racecraft, mental game

Wave 2 Modlist

Data & Telemetry

- [Assetto Corsa Telemetrick](#) — Telemetry analysis mod. **Dependencies: Content Manager.** For MoTeC i2 Pro, download from [MoTeC's official site](#). **Impact: Data analysis tool. No gameplay change.**
- [CMRT Essential HUD](#) — Real-time HUD with delta timing, radar, flags, and track limits. **Dependencies: Content Manager. Note: Replaces the old Car Track Delta app.**
- [RaceLab Apps](#) — External overlay with delta timing, standings, and weather. **Dependencies: Standalone app.**
- [Sim Racing Telemetry](#) — Lightweight telemetry viewer (free on Steam). **Dependencies: Content Manager.**

Race Practice Tools

- [AI Improvement](#) — Enhanced AI racing line consistency and aggression. **Dependencies: Content Manager.**
- [AC Grid Manager](#) — Practice standing starts with custom grid formation. **Dependencies: Content Manager. Note: Replaces the old RacePractice mod which is no longer available on Over-Take.**

Content Mods (Cars)

- [RSS GT Heart of Racing Pack](#) — GT3 and GT4 car pack (2025 version). **Dependencies: None.**
- [Audi R8 LMS 2018](#) — GT3-class Audi by TECE. **Dependencies: None.**

Content Mods (Tracks)

- [Silverstone Circuit](#) — 5.9 km, 18 turns. FIA Grade 1. **Dependencies: None.**
- [Monza](#) — 5.8 km, 11 turns. Temple of Speed. **Dependencies: None.**

Graphics (see also Wave 1 Graphics)

- Pure, C13 Aegis, A3PP, VAO Packs, and other photorealism mods are covered in the **Wave 1 Graphics** section. If you skipped Pure in Wave 1, install it here before advanced telemetry sessions.

Installation Sequence

1. Ensure Wave 1 is fully installed and verified
2. Install telemetry and data tools first (they integrate with CSP)
3. Install all content mods in any order
4. Configure MoTeC plugin in CM: **Settings** → **CSP** → **General** → **Enable telemetry logging**
5. If using Pure, verify it is active as the weather controller

Graphics Mods

This section catalogs all graphics-enhancing mods in Assetto Maximus. Graphics mods are available from Wave 1 onward.

Wave 1 Graphics

Weather & Lighting

- [Sol](#) — Comprehensive weather and lighting simulation. Replaces the default time-of-day with a dynamic, physically-based system. Features: cloud coverage simulation, volumetric fog, dynamic exposure, realistic sun positioning. **Dependencies: Custom Shaders Patch. Impact: Major visual overhaul — weather, lighting, and atmosphere are completely transformed. System: Weather/lighting engine replacement.**
- [Custom Shaders Patch](#) — Framework that enables Sol and other visual mods. Adds per-pixel lighting, dynamic shadows, rain effects, and post-processing. **Impact: Foundational graphics framework.** Use the latest stable build compatible with AC v1.16.4.

Pure (Sol Successor)

- [Pure](#) — Advanced lighting, weather, and sky controller. Successor to Sol with physically-based sky rendering, volumetric clouds, dynamic exposure, and realistic sun positioning. Delivers the most photorealistic lighting available for AC. **Dependencies: CSP 0.2.0+ . Impact: Lighting framework — replaces Sol's sky/weather model with a more advanced one. System: Weather/Lighting engine.**

Pure vs Sol: Pure offers more realistic sky colors, cloud scattering, and light dispersion. For maximum photorealism, use Pure as the primary weather controller. Pure can coexist with Sol's weather plan files but should be the active controller.

Post-Processing Filters

Post-processing filters (PP filters) control color grading, tone mapping, and bloom — the final image "look." The right PP filter transforms AC's visuals from dated to photorealistic.

- [Revolution PP Filter](#) — Clean, neutral post-processing. **Dependencies: CSP, Pure. Impact: Visual tone mapping and color grading. Note: Replaces the old Reasonable PP Filter which is no longer available on OverTake.gg.**
- [C13 Aegis PPFilter](#) — Photorealistic post-processing filter. Designed for Pure/CSP. Features realistic color science, natural highlights, deep contrast. One of the top choices for photorealism. **Dependencies: CSP, Pure recommended.**
- [A3PP \(A3 Post Processing\)](#) — Dramatic, film-style PP filter. Warm tones, cinematic contrast, natural bloom. Popular for screenshots and racing. **Dependencies: CSP.**
- [ILCB Photo Realistic Filter](#) — PP filter specifically focused on photographic realism. Neutral color palette, subtle bloom, natural gamma. **Dependencies: CSP.**

- [Natural PP Filter](#) — PP filter designed for accuracy. Minimal artistic bias, clean highlights, natural shadow detail. **Dependencies: CSP.**

Choosing a PP filter: Try 2—3 filters and pick the one that looks best on your monitor. Filters affect different monitors differently based on panel type (IPS, VA, OLED) and calibration.

Texture & Lighting Enhancements

- **4K Skin & Texture Packs** — High-resolution car skins and track textures. Search [OverTake AC Skins](#) or browse specific packs: Porsche 911 RSR ([WEC 2019](#), [Le Mans 2020](#)), [Audi RS5 DTM](#), [RSS Formula Hybrid Tyres](#). **Dependencies: Content Manager.**

Optional: Reshade

- [Reshade](#) — External post-processing injector. Adds ambient light, bloom, SMAA, LUT-based color grading, and dozens of other effects on top of AC/CSP output. **Dependencies: Standalone (injects into AC process). Impact: Additional post-processing layer. Can conflict with some CSP features — test per-filter.**

Note: Reshade is powerful but adds GPU overhead (5—15% frame time impact). Many users find Pure + C13 Aegis sufficient without Reshade.

Photorealism Checklist

To achieve a photorealistic look:

1. Install CSP (foundation)
2. Install Pure (lighting/weather)
3. Install a PP filter — C13 Aegis or A3PP recommended
4. Track textures: ensure tracks have hi-res surface textures
5. Configure CSP Lighting FX: enable dynamic exposure, high-res reflections
6. Configure Sol/Pure weather controller for your region and time of day
7. (Optional) Reshade for final color grading if desired
8. Test PP filter selection — visual preference is subjective

Content Mods

This section catalogs all car and track mods in Assetto Maximus. Content mods are available from Wave 1 onward.

Mod entries follow this format: **Name** — Description. **Dependencies**. **Compatibility notes**.

Cars

Wave 1 Cars

- [Mazda MX-5 Cup 2016](#) — Spec racing series car. Perfect for learning racecraft. **Dependencies: None.**
- [Ferrari 488 GTE](#) — GTE/GTLM endurance racer. Community conversion based on technical documentation. **Dependencies: None.**

Car Mod Requirements

All car mods must:

- Be in `.zip`, `.rar`, or `.7z` format
- Contain a `ui_car.json` or be recognized by CM
- Place files in `assetto corsa/content/cars/[car-folder]/`
- Work with AC v1.16.4 physics engine

Tracks

Wave 1 Tracks

- [Mount Panorama Bathurst](#) — 6.2 km, 23 turns. Public road turned race track. Features the iconic Conrod Straight and terrifying downhill section. **Dependencies: None.**
- [Nürburgring Nordschleife](#) — 20.8 km, 154 turns. The ultimate driver's challenge. **Dependencies: None. Note: This is a community conversion; verify compatibility with AC v1.16.4.**
- [Spa-Francorchamps](#) — 7.0 km, 19 turns. 2022 layout update. **Dependencies: None.**
- [Brands Hatch](#) — 3.9 km, 11 turns. Classic British circuit. **Dependencies: None.**
- [Silverstone Circuit](#) — 5.9 km, 18 turns. FIA Grade 1. 2024 layout. **Dependencies: None.**
- [Monza](#) — 5.8 km, 11 turns. Temple of Speed. 2022 update requires CSP. **Dependencies: CSP.**

Track Mod Requirements

All track mods must:

- Contain valid surfaces and AI lines

- Be compatible with AC v1.16.4 track structure
- Include at least one pit box layout
- Install to `assettocorsa/content/tracks/[track-folder]/`

Career & Campaign

Assetto Corsa's built-in career mode is limited to a few predefined series. These mods expand or replace it with deeper progression systems.

Career Mode Mods

Assetto Corsa does not have a dedicated career mode equivalent to games like Forza or Gran Turismo. Instead, the community uses several approaches:

- [OverTake Career Mode Tag](#) — Browse community career mode mods and tools on OverTake.gg tagged with "career-mode".

Content Manager Championships

Content Manager includes a built-in **Championship** mode (CM → Championship) that lets you create custom multi-race series with:

- Custom car and track selection
- Points systems (F1-style, GT-style, or custom)
- Practice / Qualifying / Race sessions
- AI opponents at configurable skill levels
- Weather progression across rounds

No additional mod needed — this is a CM feature that many players overlook.

Wave Placement

- **Career mods:** Search overtake.gg/tags/career-mode for available options. Install after Wave 1 (requires CSP).
- **CM Championships:** Available immediately from Wave 0 — no mod installation required.

Mechanics Mods

This section catalogs mods that alter or enhance physics, force feedback, and driving mechanics. These mods are available from Wave 1 onward and **never** in Wave 0.

Physics Mods

- [FFB Clip](#) — Analyzes force feedback output and prevents clipping (signal saturation). Provides real-time monitoring and auto-adjustment of FFB gain. **Dependencies: Content Manager System/Mechanic** **Impact: Force feedback system enhancement. Does not change physics, but improves wheel feel and prevents signal distortion. Configure via FFB Clip UI or CM settings.**
- [Minimalista HUD](#) — Clean data HUD with tyre temps, brake bias, fuel, and delta times. **Dependencies: Content Manager.** **Impact: Information display. No physics changes.** **Note: Replaces the old RST Minimal UI.**

Tyre & Handling Mods

Future waves may include dedicated tyre models (e.g., Tyre RF Pro) or handling overlays. For Wave 1, the stock AC tyre model with CSP's enhanced tyre FX is sufficient.

FFB & Wheel Settings

Detailed force feedback configuration can be found in the **Configuration** chapter.

Mod Compatibility Notes

- Mechanics mods must be tested together for conflicts
- FFB Clip works alongside all other mods
- HUD replacements like RST Minimal UI may conflict with other HUD mods — enable only one at a time
- No mechanics mod in Assetto Maximus modifies car performance data directly (no “better physics” cheat mods)

UI & Quality of Life Mods

This section catalogs all UI, HUD, and quality-of-life mods. These are the foundation of Wave 0 and remain active through all later waves.

Wave 0 (Core)

These mods are part of the Wave 0 installation and should be installed before any other mods.

Spotter & Audio

- [Sidekick](#) — Audible spotter that calls out: car proximity (left/right), gap distances, yellow/blue/checkered flags, pit lane entry/exit reminders. **Dependencies: Content Manager Impact: Audio. No visual overlay. No gameplay change. Reduces cognitive load by providing critical race information verbally. Configuration: Volume, language, and verbosity settings available in-app.**
- [Crew Chief](#) — External application providing: spotter calls, pit strategy advice, fuel calculations, tyre temperature readings, weather reports. **Dependencies: Standalone app (Windows). Runs alongside Assetto Corsa. Impact: External tool. Communicates via AC shared memory. No in-game modification. Setup: Install, launch before AC, select AC as the game. Voice recognition supports common commands.**

Visual Overlays

- [Car Radar](#) — 360° radar display positioned in the corner of the screen. Shows all nearby cars with distance and relative speed indicators. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change. Essential for close-quarters racing awareness. Configuration: Radar range, size, opacity, and position.**
- [Helicorsa](#) — 3D arrow indicator drawn in-world above nearby cars. Arrow color and size indicate proximity. Green = safe distance, Yellow = close, Red = overlapping. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change. Configuration: Trigger distance thresholds, arrow size, and enabled modes.**

Wave 1 Additions

- [Minimalista HUD](#) — Clean data HUD overlay. Replaces the old RST Minimal UI. **See 07-modlist-mechanics for full details.**
- [3D Shift Light](#) — Visual gear indicator and shift light that displays optimal shift points. **Dependencies: Content Manager Impact: Visual overlay. No gameplay change.**

UI Mod Conflicts & Rules

- Do not enable two HUD-replacement overlays simultaneously
- Sidekick and Crew Chief can run together safely (Crew Chief defers to Sidekick for spotter calls when configured)
- Car Radar and Helicorsa complement each other — running both is recommended
- All Wave 0 UI mods are compatible with all later wave mods

Configuration Guides

Graphics Settings

Video Settings (in-game)

Setting	Recommended Value	Notes
Resolution	Native monitor resolution	Match your display
VSync	Off	Reduces input lag
Fullscreen	On	Better performance
Anti-aliasing	4x MSAA	Balance quality/performance
Anisotropic Filtering	16x	Minimal performance cost
World Detail	High	
Reflection Quality	Low to Medium	Heavy performance impact at High
Shadow Quality	High	
Smoke Generation	Low	Minimal visual benefit
Post Processing	Ultra	When using CSP, managed by PP filter

Content Manager Settings

Settings → Content Manager

- **Appearance:** Dark theme recommended for readability
- **Language:** Your preference
- **Start minimized:** Optional
- **Close to tray:** Recommended to keep CM accessible

Settings → Assetto Corsa

- **Game path:** Verify it points to your AC installation
- **Launch with CSP:** Enable after Wave 1 CSP installation
- **Audio:** Enable "Use custom audio" if you install sound mods

Force Feedback Settings

Base FFB Settings (in-game)

Setting	Recommended Value	Notes
Gain	80-100%	Adjust per car
Filter	0	No smoothing
Minimum Force	0%	Avoid artificial effects
Kerb Effects	30-50%	Personal preference
Road Effects	30-50%	Personal preference
Slip Effects	50%	Helps detect loss of grip
ABS Effects	100%	If car has ABS
Gamma	1.00	Linear response

FFB Clip Configuration

1. Launch FFB Clip before AC
2. Select your wheel from the dropdown
3. Set target clipping to 90-95%
4. Enable auto-gain adjustment
5. In-game, run 3-4 laps
6. FFB Clip will suggest gain adjustment — accept it
7. Verify: FFB bar should stay mostly green, with occasional yellow spikes

Wheel-Specific Notes

Logitech G27/G29/G920

- Logitech Gaming Software: Enable "Allow game to adjust settings"
- Overall effects strength: 100%
- Spring and damper effects: 0%
- Enable centering spring: OFF

Thrustmaster (T300/TX/T150)

- Control Panel: Set rotation to 900° (or match in-game)
- Master gain: 75-85%
- No other adjustments needed

Fanatec (CSL/DD)

- Wheel Base: SEN = Auto, FF = 100, FEI = 100
- In-game: Start with lower gain (60-70%) and increase

Content Manager Advanced Configuration

Custom Shaders Patch Settings

Accessed via CM → Settings → Custom Shaders Patch

Tab	Key Settings
General	Enable "Use 64-bit AC", "Skip intro logos"

Tab	Key Settings
Lighting FX	Adjust sun, ambient, and contrast to preference
Weather FX	Enable "Sol weather script" after Sol installation
Particles FX	Enable smoke and rain particles
Smart Mirror	Set FPS limit to 30 to save performance

Sol Configuration

After installing Sol:

1. Set in-game time to 14:00
2. Weather: Select "Sol" weather plan
3. Enable "Time progression" for dynamic day/night
4. Set time multiplier to 1x (real time) or 2-5x for accelerated
5. Adjust cloud coverage in the Sol Weather Controller app

Audio Configuration

- **Settings → Audio**
- Enable "Reduce CPU load from audio" if experiencing stutters
- Set "Audio renderer" to default (Windows Audio)
- Adjust engine volume and environment volume to preference

Control Mapping

Create a profile for each control method:

1. **Settings → Controls → New Preset**
2. Name: e.g., "G29 - Default" or "Xbox Controller"
3. Map all controls precisely
4. Set steering deadzone: 0-3% (wheel) or 5-10% (gamepad)
5. Set brake/gas deadzone: 0-5%
6. Save and assign

Wheel Profiles

- **Logitech G27/G29/G920** — Overall 100%, spring/damper 0%, centering spring OFF
- **Thrustmaster T300/TX/T150** — Rotation 900°, master gain 75-85%
- **Fanatec CSL/DD** — SEN = Auto, FF = 100, FEI = 100, start gain 60-70%

Gamepad Profiles

- **Steering Speed:** 50—70% — lower values feel more stable, higher values respond faster
- **Steering Filter:** 1—5 — smooths stick input; higher = smoother but slower
- **Steering Deadzone:** 5—10% — prevents drift from worn sticks
- **Gamma:** 1.00 — linear response curve

- **Speed Sensitivity:** 0% — disables speed-dependent steering reduction
- **Vibration:** Enable for tyre slip and kerb feedback
- **Throttle Saturation:** 90—95% — leaves room for modulation at the top of the trigger range

Performance Optimization

Low-End Systems

- Reduce World Detail to Medium
- Disable mirrors (or use Smart Mirror with 15 FPS limit)
- Shadow Quality: Low
- Disable Post Processing
- Run in 1080p
- Close background applications

High-End Systems

- World Detail: Maximum
- Reflection Quality: High
- Shadow Quality: Maximum
- Post Processing: Ultra
- Enable all CSP visual features
- Use 4K resolution if supported

Appendix

Troubleshooting

Game Won't Launch

1. Verify game files in Steam (Properties → Installed Files → Verify integrity)
2. Uninstall/reinstall Content Manager
3. Delete `Documents/Assetto Corsa/cfg/` and let AC regenerate configs
4. Disable all mods in CM → launch stock AC → re-enable mods one by one

Content Manager Fails to Detect AC

1. Manually set game path in CM Settings → Assetto Corsa → Game path
2. Ensure Steam is running
3. Check that you launched AC at least once (creates registry keys CM reads)

CSP Not Working

1. Ensure CSP is installed in `assettocorsa/extensions/`
2. Check that you launched AC through CM (not Steam directly)
3. CM Settings → Assetto Corsa → enable "Use 64-bit AC"
4. Verify CSP version matches your AC version

Mods Not Appearing

1. Check the mod is installed (CM → Content tab)
2. Ensure the mod is enabled (checkbox next to it)
3. For cars: check `assettocorsa/content/cars/` for a valid folder
4. For tracks: check `assettocorsa/content/tracks/`
5. Restart CM

Performance Issues After Mod Installation

1. Lower CSP settings (especially Reflections and Shadows)
2. Disable unnecessary overlays
3. Reduce mirror resolution in CSP
4. Check for conflicting mods (see `conflicts.md`)

Force Feedback Issues

1. Verify wheel drivers are installed
2. Check that AC is selected as the game in FFB Clip

3. Reduce overall FFB gain to 70% and gradually increase
4. Set "Filter" to 0 in AC FFB settings
5. Disable "Minimum Force" and "Spring/Damper" effects in wheel software

Performance Reference

Component	Minimum	Recommended	Ideal
CPU	Intel i5-2500K / AMD FX-8150	Intel i7-4790K / AMD Ryzen 5 3600	Intel i7-10700K / AMD Ryzen 7 5800X
GPU	NVIDIA GTX 660 / AMD R9 270	NVIDIA GTX 1070 / AMD RX 580	NVIDIA RTX 3070 / AMD RX 6800
RAM	8 GB	16 GB	32 GB
Storage	HDD	SSD	NVMe SSD

Glossary

Apex — The innermost point of a racing line through a corner.

Camber — The vertical tilt of a wheel relative to the road surface. Negative camber improves cornering grip.

Content Manager (CM) — Third-party launcher and mod manager for Assetto Corsa. The primary tool for this guide.

CSP (Custom Shaders Patch) — A community-created patch for Assetto Corsa that adds advanced graphics, weather, and post-processing effects.

FFB (Force Feedback) — Haptic feedback from the steering wheel that communicates tyre grip, road surface, and car behavior.

FOV (Field of View) — The extent of the observable game world visible at any moment. Correct FOV is critical for accurate depth perception.

Oversteer — Condition where the rear wheels lose grip, causing the car to rotate more than intended.

PP Filter (Post-Processing Filter) — A visual filter applied after rendering that affects color, contrast, bloom, and tone mapping.

Sol — A weather simulation system for Assetto Corsa that requires CSP. Adds dynamic time-of-day and realistic weather.

Understeer — Condition where the front wheels lose grip, causing the car to continue straight despite steering input.

Useful Links

- OverTake.gg — Assetto Corsa Mods

- [Content Manager Official Site](#)
- [Assetto Corsa Modding Guide](#)
- [r/assettocorsa on Reddit](#)
- [OverTake.gg — AC Mod Database](#)

Migration Note: RaceDepartment → OverTake.gg

RaceDepartment was acquired by OverTake.gg. All mod download URLs now use the `overtake.gg` domain. Most mod IDs were reassigned during the migration, so links from earlier versions of this guide may need updating. If a mod URL returns a 404, search for the mod by name on [OverTake.gg](#).

Credits

- **Author:** The Assetto Maximus team
- **Logo:** Assetto Maximus brand assets
- **Mod Credits:** Each mod's respective author(s) — see individual mod pages for full credits
- **Special Thanks:** The Assetto Corsa modding community

Assetto Corsa is a product of Kunos Simulazioni. This guide is a community project and is not affiliated with or endorsed by Kunos Simulazioni.

